

Md. Tanvir Ahmed Tonmoy

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Professional Summary

Electrical and Electronic Engineering undergraduate with hands-on experience in RTL design, VHDL/Verilog, FPGA-oriented digital design, processor architecture, finite-state machines, testbench development, waveform-based verification, and CMOS/VLSI design. Currently designing a simplified RISC-V processor in synthesizable VHDL and an SRAM-based in-memory computing architecture in Xilinx ISE. Experienced with Cadence Virtuoso, Genus, Innovus, Python, and embedded hardware development.

Education

BRAC University, Dhaka, Bangladesh

Bachelor of Science in Electrical and Electronic Engineering
Specialization: Electronics

Expected September 2026

CGPA: **3.68/4.00**

Technical Skills

HDL / RTL

VHDL, Verilog, RTL Design, FSM Design, Synthesizable Design, Testbench Development, Functional Simulation

Digital / VLSI

RISC-V Processor Design, SRAM Architecture, Compute-in-Memory

Programming

Python, C, C++, MATLAB

FPGA / EDA

Xilinx ISE, Cadence Virtuoso, Genus, Innovus, Maestro, Waveform Analysis, Timing Verification

Embedded Systems

ESP32, STM32, Arduino, Teensy, Communication protocols, Wireless Modules

Tools

GitHub, VS Code, LaTeX, Altium Designer, Autodesk Eagle, Proteus, PSPICE

Relevant RTL, FPGA & VLSI Projects

VHDL-FPGA Based RISC-V Processor Design

2026 – Present

- Designing and implementing a simplified RISC-V processor using synthesizable VHDL for FPGA implementation.
- Developing modular processor components and integrating them into a complete processor architecture.
- Building a comprehensive VHDL testbench to verify supported instruction execution through simulation and waveform analysis.
- Preparing the design for synthesis, functional demonstration, and FPGA-oriented implementation.

SRAM-Based In-Memory Computing Architecture in VHDL

2026

- Designed and simulated an SRAM-based in-memory computing architecture using VHDL in Xilinx ISE.
- Developed memory, computational, and control logic with an FSM to coordinate memory and in-memory compute operations.
- Verified functionality using VHDL testbenches, simulation waveforms, and synthesized technology schematics.

Variation-Aware SRAM Compute-in-Memory Architecture

2026

- Designed and simulated a 45 nm 6T SRAM-based compute-in-memory Boolean architecture in Cadence Virtuoso.
- Implemented AND, OR, NAND, NOR, XOR, and XNOR operations and evaluated transient behavior.
- Performed Monte Carlo timing analysis to quantify timing-induced logic errors under process variation.
- Developed a Python/PyTorch hardware-aware BNN evaluation flow linking circuit-level error probability with inference accuracy.

8T SRAM Crossbar and SRAM-Based Spiking Neural Network

2026 – Present

- Designing and characterizing an 8T SRAM cell in 45 nm CMOS and developing a 4×4 SRAM crossbar.
- Extending the architecture toward an SRAM-based spiking neural network for neuromorphic computing applications.

45 nm CMOS Phase-Locked Loop

2026

- Designed a CMOS PLL in Cadence Virtuoso including PFD, charge pump, loop filter, VCO, and divide-by-48 frequency divider.

- Performed DC/transient analysis and evaluated lock behavior, frequency response, and current consumption.

Engineering Experience

Industrial Attachment – Neural Semiconductor Ltd.

2026

Exposure to semiconductor design workflows including digital design, physical design, analog design, RF design, and fabrication-related processes.

Electronics Subsystem Lead – BRACU Diganta

2024 – Present

Led onboard electronics design, sensor integration, telemetry hardware, power planning, and subsystem integration for model-satellite systems. Worked on CanSat 2025 hardware development; the team was selected as a global finalist in Virginia, USA.

Undergraduate Teaching Assistant – BRAC University

February 2025 - June 2025

Supported undergraduate Electrical and Electronic Engineering coursework and student learning activities.

Selected Publications

- H. H. Jui, **M. T. A. Tonmoy**, et al., “Variation-Aware SRAM Compute-in-Memory Boolean Architecture for Binary Neural Networks,” *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2026, Kolkata, India.
- F. Uddin, N. Shahrair, M. M. Rahman, **M. T. A. Tonmoy**, et al., “Soft Robotic Bellow-Controlled Arm Using Multi-DOF Control System,” *INTCEC 2025*, Chicago, USA.
- M. M. Chowdhury, M. M. Rahman, M. Mahmud, K. A. Rahim, **M. T. A. Tonmoy**, et al., “BRACU Diganta: An Affordable CanSat Learning Kit for Students in Developing Countries,” *International Astronautical Congress (IAC/IAF)*, 2025, Sydney, Australia.

Selected Achievements

- Dean’s List for six semesters.
- Merit-based tuition scholarship: 25% for four semesters and 10% for three semesters.
- CanSat Competition 2025 Global Finalist, American Astronautical Society, Virginia, USA.

References

- **Twisha Titirsha, PhD**

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